

# EN.553.740 Machine Learning 1

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## 0.1 Introduction

A good way to think of machine learning is instead of a computer taking data and a program to produce output, it takes data and output and attempts to produce a program, ie "learn" a function.

We use it for tasks that are too complex to program, like driving or large scale data analysis. A bit of history of "learning":

- 1959: Arthur Samuel (IBM) created a program to play checkers, introducing rule-based learning.
- 1960: Frank Rosenblatt introduced a linear classifier based on brain neurons, called a Perceptron.
- 1969: Marvin Minsky showed limitations of the perceptron, halting development for 20 years
- 1980: First workshop in machine learning
- 1986: Multilayer networks and backpropagation introduced
- 1990: Vapnik and Chervonenkis introduce kernels and SVM, and termed "Statistical Learning"
- 2002: Rise of graphical models
- 2010: Deep learning takes over
- 2016: ML hits record size with NIPS and ICML doubling in size

Machine learning has been used for a variety of tasks, like stock market prediction, decoding thoughts from brain scans, and self-driving cars.

### 0.1.1 Learning Problems

Statistical Learning falls under five rough fields:

- Classification (Assign a category)
- Regression (predict a real value)
- Ranking (order items)
- Clustering (item partitioning, like image segmentation)
- Dimensionality Reduction/Manifold Learning (Computer vision)

In a broader sense, there is supervised learning (Classification/regression) and unsupervised learning (Clustering, dimensionality reduction, density estimation) and things like semi-supervised learning, active learning, reinforcement learning, etc.

**Supervised Learning** For a given problem, we learn a function  $f : X \rightarrow Y$  such that  $x \in \mathcal{X}$ ,  $y \in \mathcal{Y}$ . Classification deals with discrete labels while regression deals with continuous functions.

**Unsupervised Learning** For a given problem, we learn a function  $f : X \rightarrow Y$  such that  $x \in \mathcal{X}$ ,  $f(x) \in \mathbb{R}$ . Usually embedding (for text), clustering, or density estimation.

## 0.2 Mathematical Review

Algorithms are built from sets, functions, vectors, matrices, norms, and derivatives, so we introduce common notation here. Consider a student success dataset with the target being quiz score, and features as study time and prior score.

### 0.2.1 Linear Algebra

#### 0.2.1.1 Sets and Power Sets

Let  $A = \{1, 2, 3, 4, 5\}$  be the set of students in the toy dataset. The *power set* is  $\mathcal{P}(A) = \{B : B \subseteq A\}$ . If  $|A| < \infty$ , then  $|\mathcal{P}(A)| = 2^{|A|}$ . A subset  $B \subseteq A$  may represent a mini-batch, validation set, or a selected group of samples.

#### 0.2.1.2 Functions

For sets  $A, B$ , write  $B^A = \{f : A \rightarrow B\}$ . if  $B = \mathbb{R}$ , then  $\mathbb{R}^A$  is the set of real-valued functions on  $A$ . If  $|A| < \infty$ , then  $\mathbb{R}^A \simeq \mathbb{R}^{|A|}$  as vector spaces. For our toy example:

$$x_1 : A \rightarrow \mathbb{R} \quad x_2 : A \rightarrow \mathbb{R} \quad y : A \rightarrow \mathbb{R}$$

For a subset  $C \subseteq A$ , define an *indicator function*:

$$\mathbb{1}_C : A \rightarrow \{0, 1\}, \quad \mathbb{1}_C(i) = \begin{cases} 1, & i \in C, \\ 0, & i \notin C \end{cases}$$

Indicators are used to encode events, groups, labels, masks, etc.

### 0.2.1.3 Vectors

We use a column convention for vectors. For each student  $i$ , write the feature vector as:

$$x_i = \begin{bmatrix} x_i^{(1)} \\ x_i^{(2)} \end{bmatrix} \in \mathbb{R}^2$$

Standard vectors that are useful include the zero vector ( $0_d$ ), unit vector ( $\mathbb{1}_d$ ), and identity matrix ( $I_{\mathbb{R}^d}$ ):

$$0_d = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \quad \mathbb{1}_d = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \quad I_{\mathbb{R}^d} = \begin{bmatrix} 1 & \cdots & 0 \\ & \ddots & \\ 0 & \cdots & 1 \end{bmatrix}$$

### 0.2.1.4 $\ell^p$ norms

For  $p \geq 1$  and  $x \in \mathbb{R}^d$ , a  $p$ -norm is defined as

$$\|x\|_p = (|x^{(1)}|^p + \cdots + |x^{(d)}|^p)^{\frac{1}{p}}$$

. Special cases include:

$$\ell_1 = \sum_j |x^{(j)}| \quad \ell_2 = \sqrt{\sum_j (x^{(j)})^2} \quad \ell_\infty = \max_j |x^{(j)}|$$

$\ell^2$  is called the euclidean norm. We require  $p \geq 1$  because the triangle inequality holding is essential for a norm, and only holds for  $p \geq 1$ :

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p$$

$$\text{when } p = 1; |x| + |y| \leq 1$$

$$\text{when } p = \infty; \max\{|x|, |y|\} \leq 1 \iff |x| \leq 1,$$

A good visualization of norms is shown below:

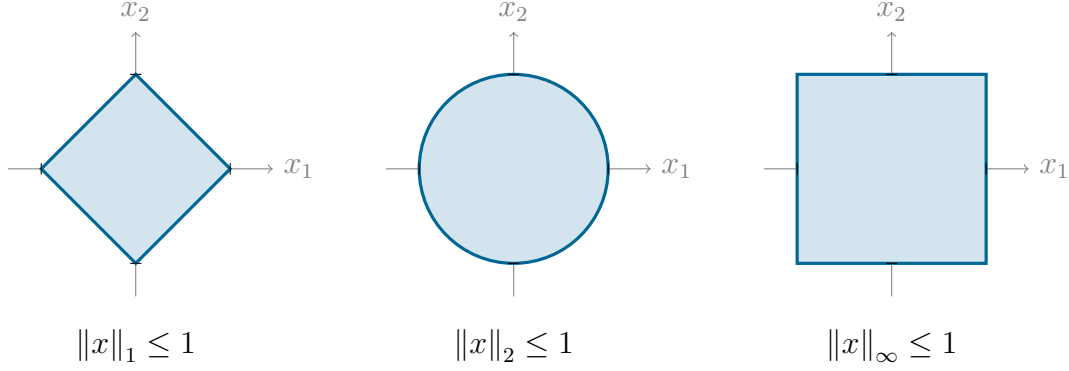


Figure 1: Unit balls in  $\mathbb{R}^2$ . Their boundaries consist of vectors with norm equal to one.

**Operator Norms** For  $A \in \mathcal{M}_{m,d}$ ,  $\|A\|_{op} := \max\{\|Ax\|_2 : x \in \mathbb{R}^d, \|x\|_2 = 1\}$ .  $\|A\|_{op}$  is the largest stretching factor of the linear map  $x \mapsto Ax$ . Alternatively, we can derive a similar definition:

$$\begin{aligned}\|Ax\|_2^2 &= \langle Ax, Ax \rangle \\ \langle Ax, Ax \rangle &= (Ax)^T Ax = x^T A^T Ax \\ \|A\|_{op}^2 &= \max_{\|x\|_2=1} x^T A^T Ax\end{aligned}$$

Operator norms quantify how much perturbations can be amplified.

### 0.2.1.5 Cauchy-Schwarz Inequality

For  $x, y \in \mathbb{R}^d$ ,  $|\langle x, y \rangle| \leq \|x\|_p \|y\|_q$  with equality iff  $x = \alpha y$  for some scalar  $\alpha$  and  $\frac{1}{p} + \frac{1}{q} = 1$ .

**Proof sketch:** For any  $\alpha \in \mathbb{R}$ ,

$$0 \leq \|x - \alpha y\|_2^2 = \|x\|_2^2 - 2\alpha \langle x, y \rangle + \alpha^2 \|y\|_2^2$$

Choosing the best  $\alpha$  gives the inequality:

$$\begin{aligned}\forall \alpha \in \mathbb{R}, \quad & \|x - \alpha y\|_2^2 \geq 0 \\ \iff & \langle x - \alpha y, x - \alpha y \rangle \geq 0 \\ \iff & \langle x, x \rangle - 2\alpha \langle x, y \rangle + \alpha^2 \langle y, y \rangle \geq 0 \\ \iff & \|x\|_2^2 - 2\alpha \langle x, y \rangle + \alpha^2 \|y\|_2^2 \geq 0\end{aligned}$$

Without loss of generality, assume  $\|y\|_2 \geq 0$  and take  $\alpha = \frac{\langle x, y \rangle}{\|y\|_2^2}$ :

$$\begin{aligned}\implies & \|x\|_2^2 - \frac{2 \langle x, y \rangle \|x\|_2}{\|y\|_2} + \|x\|_2 \geq 0 \\ & \|x\|_2 \|y\|_2 \geq \langle x, y \rangle\end{aligned}$$

Equality means  $x - \alpha^* y = 0$  so  $x = \alpha^* y$ .

Cauchy-Schwarz is used in ML to control linear scores, since most models are about estimating weights. A linear model estimates  $f_w(x) = \langle w, x \rangle$ . Cauchy-Schwarz gives

$$|f_w(x)| = |\langle w, x \rangle| \leq \|w\|_2 \|x\|_2$$

This is saying if features have a bounded norm, and weights have a bounded norm, scores cannot explode. This is why norm constraints and regularization appear throughout learning problems.

The inner product cannot be larger than length times length

We now look at a particle sensor example. Suppose independent sensors  $X_1, X_2, \dots, X_n$  estimate the same mean  $\mu$ :

$$\mathbb{E}(X_i) = \mu \quad \text{Var}(X_i) = \sigma_i^2$$

Among weighted averages

$$\hat{\mu} = \sum_{i=1}^n w_i X_i$$

which weights should we use, keeping  $\hat{\mu}$  unbiased? Compute:

$$\mathbb{E}(\hat{\mu}) = \mathbb{E}\left[\sum_i w_i X_i\right] = \sum_i w_i \mathbb{E}X_i = \sum_i w_i \mu = \mu$$

Thus

$$\mathbb{E}(\hat{\mu}) = \mu \iff \sum_i w_i = 1$$

Because  $X_i$  are independent,

$$\text{Var}(\hat{\mu}) = \text{Var}\left(\sum_i w_i X_i\right) = \sum_i w_i^2 \text{Var}(X_i) = \sum_i w_i^2 \sigma_i^2$$

The optimization problem we obtain is therefor minimize  $\text{Var}(\hat{\mu})$  s.t.  $\sum_i w_i = 1$  since the sum of weights is one,

$$1 = \left(\sum_i w_i\right)^2 = \left(\sum_i (w_i \sigma_i) \frac{1}{\sigma_i}\right)^2$$

Now by Cauchy-Schwarz,

$$1 = \left(\sum_i w_i\right)^2 \leq \left(\sum_i w_i^2 \sigma_i^2\right) \left(\sum_i \frac{1}{\sigma_i^2}\right)$$

Recall  $|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2$ . Therefore

$$\text{Var}(\hat{\mu}) = \sum_i w_i^2 \sigma_i^2 \geq \left(\sum_i \sigma_i^{-2}\right)^{-1}$$

This makes it so the elements of  $X$  and  $y$  are  $w_i \sigma_i$  and  $1/\sigma_i$  respectively.

Equality in Cauchy-Schwarz holds when

$$w_i \sigma_i \propto \sigma_i^{-1} \implies w_i \propto \frac{1}{\sigma_i^2}$$

so we should normalize weights so they sum to one:

$$\frac{\frac{1}{\sigma_i^2}}{\sum_{j=1}^n \frac{1}{\sigma_j^2}}$$

meaning lower variance sensor implies larger weight.

### 0.2.1.6 Sparsity and Lasso

Sparsity is roughly defined as what happens when  $p \rightarrow 0$ . For a nonzero scalar  $a$ :

$$\lim_{p \rightarrow 0} \sum_j |x^{(j)}|^p = \sum_j \lim_{p \rightarrow 0} |x^{(j)}|^p \because \lim_{p \rightarrow 0} |a|^p = 1 \text{ ( imagine } \begin{cases} 0, x^{(j)} = 0 \\ 1, x^{(j)} \neq 0 \end{cases} )}$$

For  $a = 0$ ,  $|0|^p = 0$ . Therefore informally,

$$\lim_{p \rightarrow 0} \sum_j |x^{(j)}|^p = \{j : x^{(j)} \neq 0\} = \|x\|_0$$

The quantity  $\|x\|_0$  counts the number of nonzero entries, but it is not a norm and optimization with it is often difficult. Common times in ML, we replace  $\|x\|_0$  with  $\|x\|_1$  to encourage sparsity while keeping convexity

### 0.2.1.7 Matrices

In ML, rows often represent samples and columns represent features. we say  $\mathcal{M}_{m,d} = \mathcal{M}_{m,d}(\mathbb{R}) = \{m \times d \text{ real matrices}\}$ . Additionally  $\mathcal{M}_d = \mathcal{M}_{d,d}$ . Notation may differ, so remember its always (across, down) ie (i,j)/(r,c)/(m,d)

By this definition, a linear prediction vector is  $X_w = [\langle x_1, w \rangle, \langle x_2, w \rangle, \dots, \langle x_i, w \rangle]^T$ .

**Symmetric and Positive Semidefinite Matrices** A symmetric matrix is defined as

$$S_d = \{A \in \mathcal{M}_d : A = A^T\}$$

A positive semidefinite matrix (PSD) is defined as

$$S_d^+ = \{A \in S_d : x^T A x \geq 0 \quad \forall \quad x \in \mathbb{R}^d\}$$

These usually appear as covariance and correlation matrices, and Hessians of convex quadratic objective functions. A positive definite matrix is defined as

$$S_d^{++} = \{A \in S_d : x^T A x > 0 \quad \forall \quad x \neq 0\}$$

This usually comes up when doing ridge regressions and finding hessians with a parameter greater than 0.

**Bilinear Forms** A bilinear form is a map  $a : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$  that is linear in each variable. If  $a$  is symmetric (ie  $a(x, y) = a(y, x)$ ) then  $a(x, y) = x^T A y$  for some symmetric matrix  $A \in S_d$ .

Inner products are PSD:

$$\begin{aligned} a(x, y) &= \langle x, y \rangle = x^T y \\ a(x, y) &= \langle x, y \rangle = x^T I_d y \\ a(x, x) &= x^T x = \|x\|_2^2 \geq 0 \end{aligned}$$

**Orthogonal Matrices** An Orthogonal transformation matrix  $O$  is defined as

$$O_{m,d} = \{A \in \mathcal{M}_{m,d} : AA^T = I_m\}$$

and for  $Q \in O_d$ ,  $Q^T Q = Q Q^T = I_d$ . The key idea is orthogonal transformations preserve euclidean geometry:

$$\|Qx\|_2^2 = (Qx)^T (Qx) = x^T Q^T Q x = x^T x = \|x\|_2^2$$

Similarly

$$\langle Qx, Qy \rangle = x^T Q^T Q y = \text{inner}xy$$

Rotations preserve distances and dot products, which matter for PCA, kernels, and preprocessing

### 0.2.1.8 Open and Closed Balls

We introduce some basic topology. For  $x \in \mathbb{R}^d, r > 0$ : A ball is open if

$$B(x, r) = \{y \in \mathbb{R}^d : \|y - x\|_2 < r\}$$

A ball includes the boundary if

$$\bar{B}(x, r) = \{y \in \mathbb{R}^d : \|y - x\|_2 \leq r\}$$

A set  $U \subset \mathbb{R}^d$  is open if

$$\forall x \in U, \quad \exists r > 0, \quad B(x, r) \subset U$$

A set is closed if its complement is open. The intuition is that inside an open set, every point has a small safety ball around it.

For  $A \subset \mathbb{R}^d$ , the interior ( $f(A)$ ) is the largest open set containing  $A$ , and the closure  $cl(a)$  is the smallest closed set containing  $A$ . We can also think of this as the closure including the boundary whereas the interior doesn't.

**Compactness** : A set  $K \subset \mathbb{R}^d$  is compact if every sequence of points in  $K$  contains a subsequence that converges to a point in  $K$ . Take for instance  $(0,1)$  vs  $[0,1]$ : The sequence  $x_n = n^{-1}$  lies in  $(0,1)$  and converges to 0, but  $0 \notin (0,1)$ , so  $(0,1)$  is not compact. For  $[0,1]$ , limits of convergent subsequences stay inside the set.



**Heine-Borel Theorem** A finite-dimensional Euclidean space:

$$K \subset \mathbb{R}^d \text{ compact} \iff K \text{ is closed and bounded.}$$

Bounded means  $\exists R > 0$  s.t.  $K \subset B(0, R)$ . This is special to finite dimensional euclidean spaces, not other spaces.

**Weierstrass Extreme Value Theorem** If  $K \subset \mathbb{R}^d$  is compact and  $f : K \rightarrow \mathbb{R}$  is continuous, then  $f$  attains its minimum and maximum on  $K$ . Put otherways,

$$\exists x_{min}, x_{max} \in K \text{ s.t. } f(x_{min}) = \min_{x \in K} f(x), \quad f(x_{max}) = \max_{x \in K} f(x)$$

In optimization, we often write  $\inf_{w \in K} \ell(w)$ . Weierstrass tells us when this minimum is actually achieved. Basically saying that on compact parameter sets, continuous objectives have optimizers. We look at an example:

Look at constrained regression. Consider minimizing least squares over a compact ball

$$K_R = \{w \in \mathbb{R}^d : \|w\|_2 \leq R\}$$

The objective is

$$\ell(w) = \frac{1}{2n} \|Xw - y\|_2^2$$

We can prove this by showing  $\ell(w)$  is continuous in  $w$ . Then show that  $K_R$  is closed and bounded in  $\mathbb{R}^d$ . By Heine-Borel,  $K_R$  is compact, so by Weierstrass,  $\exists \hat{w} \in K_R$  s.t.  $\ell(\hat{w}) = \min_{w \in K_R} \ell(w)$ .

### 0.2.1.9 Metric Spaces

A metric space is a set  $\mathcal{B}$  equipped with a distance  $\rho : \mathcal{B} \times \mathcal{B} \rightarrow [0, \infty)$  satisfying  $\forall x, y, z \in \mathcal{B}$ :

$$\rho(x, y) = 0 \iff x = y$$

$$\rho(x, y) = \rho(y, x)$$

$$\rho(x, z) \leq \rho(x, y) + \rho(y, z)$$

For  $\mathcal{B} \in \mathbb{R}$  and  $\rho : \mathbb{R} \times \mathbb{R} : \rho(x, y) = |x - y|$ .

For  $\mathcal{B} \in \mathbb{R}^d$  and  $\rho : \mathbb{R}^d \times \mathbb{R}^d : \rho(x, y) = \|x - y\|_2$ .

Metrics formalize dissimilarity between data points. This could be distances between two feature vectors, probability distributions, or functions/signals. Open, Closed, and compactness still make sense in metric spaces, but Heine-Borel may fail. In  $\mathbb{R}$ , closed and bounded iff compact, but in a general metric space, it does not imply compactness

## 0.2.2 Calculus

### Calculus Notation Map

| Notation              | Meaning in these notes                                                            |
|-----------------------|-----------------------------------------------------------------------------------|
| $x, y \in \mathbb{R}$ | one-dimensional points, often with $x < y$                                        |
| $x \in \mathbb{R}^d$  | deterministic vector or input point                                               |
| $e_1, \dots, e_d$     | canonical basis of $\mathbb{R}^d$ : $e_1 = (1, 0, \dots)$ , $e_2 = (0, 1, \dots)$ |
| $\partial f$          | derivative in one dimension                                                       |
| $\partial_j f$        | partial derivative along $e_j$                                                    |
| $df(x)$               | differential or Jacobian matrix at $x$                                            |
| $\nabla \cdot f$      | divergence, when $f: \mathbb{R}^d \rightarrow \mathbb{R}^d$                       |
| $\nabla^2 f$          | Hessian, for scalar-valued $f$                                                    |
| $\Delta f$            | Laplacian, $\text{tr}(\nabla^2 f)$                                                |

### Probability Notation Map

| Notation                   | Meaning in these notes                        |
|----------------------------|-----------------------------------------------|
| $\Omega$                   | sample space                                  |
| $A \subseteq \Omega$       | event                                         |
| $\mathbb{P}$               | probability function or probability measure   |
| $(\Omega, \mathbb{P})$     | probability space                             |
| $X, Y$                     | random variables (capital letters)            |
| $x, y$                     | realizations or values of random variables    |
| $R_X, R_Y$                 | possible values of $X$ and $Y$                |
| $[X = x]$                  | event $\{\omega \in \Omega : X(\omega) = x\}$ |
| $\mathbb{E}[X]$            | expectation of $X$                            |
| $\mathbb{E}[Y \mid X = x]$ | conditional expectation of $Y$ given $X = x$  |

A model is often a function  $f(x)$  and a loss is often a function  $\ell(w)$  of parameters  $w$ . Optimization therefore asks how does  $\ell(w)$  change after  $w$  is perturbed, or, if we move from  $w$  to  $w + h$ , can we predict whether loss goes down?

**Intervals** Assume  $x, y \in \mathbb{R}; x > y$ . The line segment from  $x$  to  $y$  is written by  $(1 - t)x + ty$ . Define the intervals as follows:

$$[x, y] := \{(1 - t)x + ty : 0 \leq t \leq 1\}$$

$$[x, y) := \{(1 - t)x + ty : 0 \leq t < 1\}$$

$$(x, y] := \{(1 - t)x + ty : 0 < t \leq 1\}$$

$$(x, y) := \{(1 - t)x + ty : 0 < t < 1\}$$

The same line segment formula will appear in convexity, Jensen's Inequality, and Taylor expansion along a direction.

### 0.2.2.1 1D Derivatives

Let  $f : I \subset \mathbb{R} \rightarrow \mathbb{R}$ ,  $t \mapsto f(t) \in \mathbb{R}$ . The derivative of  $f$  at  $t_0 \in I$  is

$$\partial f(t_0) = \partial f|_{t=t_0} := \lim_{h \rightarrow 0} \frac{f(t_0 + h) - f(t_0)}{h}$$

when this limit exists. For higher-order derivatives:

$$\partial^k f := \partial(\cdots \partial(\partial f)), \quad k \geq 0, \quad k \in \mathbb{Z}^+$$

with the convention  $\partial^0 f := f$ . The derivative is the local slope of the function at  $t_0$ .

### 0.2.2.2 Higher-Dimensional Derivatives

Now move to  $d > 1$ , and let  $f : U \subset \mathbb{R}^d \rightarrow \mathbb{R}^m$  and  $U$  is *open* (meaning we can move a little around  $x$  without immediately leaving the domain).

### 0.2.2.3 Partial Derivates

Let  $e_1, \dots, e_d$  be the canonical basis of  $\mathbb{R}^d$ . For  $1 \leq j \leq d$ :

$$\partial_j f(x) := \partial_t(f(x + te_j))|_{t=0} = \lim_{t \rightarrow 0} \frac{f(x + te_j) - f(x)}{t}$$

This means that  $e_j$  changes only coordinate  $j$  and  $x + te_j$  means move from  $x$  only along coordinate  $j$ . The partial derivative measures local change in that coordinate direction.

For example, let  $f(\theta) = f(x, y) = x^2 + 3xy$ . Then  $\partial_x f(\theta) = 2x + 3y$  and  $\partial_y f(x) = 3y$ , and at  $\theta = (1, 2)^T$ ,  $\partial_x f(\theta) = 8$  and  $\partial_y f(x) = 3$

### 0.2.2.4 Directional Derivatives

Partial derivatives measure change along the coordinate directions. More generally, for a direction  $h \in \mathbb{R}^d$ , the *directional derivative* of  $f$  at  $x$  in the direction  $h$  is

$$\partial_t f(x + th)|_{t=0}.$$

A partial derivative is the special case  $h = e_j$ , whereas a directional derivative allows any  $h \in \mathbb{R}^d$ .

In machine learning, if  $x$  is an input image or feature vector and  $h$  is a small perturbation, the directional derivative describes how the model output changes under that perturbation.

### 0.2.2.5 The Differential

Let  $f: U \subset \mathbb{R}^d \rightarrow \mathbb{R}^m$ , with component functions  $f^{(1)}, \dots, f^{(m)}$ . The *differential* of  $f$  at  $x$  is

$$df(x) := \left( \partial_j f^{(i)}(x) \right)_{1 \leq i \leq m, 1 \leq j \leq d}.$$

Equivalently,

$$df(x) = \begin{pmatrix} \partial_1 f^{(1)}(x) & \cdots & \partial_d f^{(1)}(x) \\ \vdots & \ddots & \vdots \\ \partial_1 f^{(m)}(x) & \cdots & \partial_d f^{(m)}(x) \end{pmatrix} \in \mathbb{R}^{m \times d}.$$

Thus, row  $i$  contains the partial derivatives of the output component  $f^{(i)}$ , while column  $j$  contains the partial derivatives with respect to the input coordinate  $x_j$ .

**Example 0.2.1** (Computing  $df(x)h$ ). Consider the scalar-valued function

$$f(x^{(1)}, x^{(2)}) = (x^{(1)})^2 + 3x^{(1)}x^{(2)}.$$

Its differential is the row vector

$$df(x) = \begin{pmatrix} \partial_1 f(x) & \partial_2 f(x) \end{pmatrix} = \begin{pmatrix} 2x^{(1)} + 3x^{(2)} & 3x^{(1)} \end{pmatrix}.$$

At  $x = (1, 2)^T$ ,

$$df(x) = \begin{pmatrix} 8 & 3 \end{pmatrix}.$$

Therefore, if  $h = (1, -1)^T$ , then

$$df(x)h = \begin{pmatrix} 8 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 5.$$

### 0.2.2.6 Product Rule

Let  $f, g: U \rightarrow \mathbb{R}$ . Then

$$d(fg)(x) = f(x) dg(x) + g(x) df(x).$$

The product rule says that the first-order change of a product has two contributions: one from changing  $f$  and one from changing  $g$ .

### 0.2.2.7 Chain Rule

Let

$$f: U \subset \mathbb{R}^d \rightarrow \mathbb{R}^m, \quad g: V \subset \mathbb{R}^b \rightarrow U.$$

The condition  $g(V) \subseteq U$  ensures that  $f(g(x))$  is defined for every  $x \in V$ . The composition satisfies  $f \circ g: V \rightarrow \mathbb{R}^m$ , and the chain rule is

$$d(f \circ g)(x) = df(g(x)) dg(x).$$

The dimensions are compatible because

$$\underbrace{df(g(x))}_{m \times d} \underbrace{dg(x)}_{d \times b} = \underbrace{d(f \circ g)(x)}_{m \times b}.$$

Modern machine-learning models are compositions of functions:

$$x \mapsto h_1(x) \mapsto h_2(h_1(x)) \mapsto \cdots \mapsto f(x).$$

The chain rule explains how derivatives propagate through such a composition. In particular, backpropagation in a neural network is repeated application of the chain rule.

### 0.2.2.8 Divergence

When  $d = m$ , a function  $f: U \subset \mathbb{R}^d \rightarrow \mathbb{R}^d$  is a vector field. Its *divergence* is

$$\nabla \cdot f(x) := \text{tr}(df(x)).$$

Writing

$$f(x) = \begin{pmatrix} f^{(1)}(x) \\ \vdots \\ f^{(d)}(x) \end{pmatrix},$$

the divergence is the sum of the diagonal entries of the  $d \times d$  differential:

$$\nabla \cdot f(x) = \partial_1 f^{(1)}(x) + \cdots + \partial_d f^{(d)}(x).$$

**Example 0.2.2** (Computing divergence). Let  $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  be

$$f(x) = \begin{pmatrix} (x^{(1)})^2 \\ x^{(1)}x^{(2)} \end{pmatrix}.$$

Then

$$df(x) = \begin{pmatrix} 2x^{(1)} & 0 \\ x^{(2)} & x^{(1)} \end{pmatrix}.$$

Therefore,

$$\nabla \cdot f(x) = \text{tr}(df(x)) = 2x^{(1)} + x^{(1)} = 3x^{(1)}.$$

Divergence appears in continuous normalizing flows, probability flows, and differential equations for densities.

### 0.2.2.9 Determinant and Its Derivatives

Let

$$A = \begin{bmatrix} a_1 & \cdots & a_d \end{bmatrix} \in \mathcal{M}_d,$$

where  $a_i$  is the  $i$ th column of  $A$ . The determinant

$$\det: \mathcal{M}_d \rightarrow \mathbb{R}$$

can be viewed as a  $d$ -linear form in the columns:

$$\delta(a_1, \dots, a_d) = \det(A).$$

It is characterized by the following properties:

- If  $a_i = a_j$  for  $i \neq j$ , then  $\delta(a_1, \dots, a_d) = 0$ .
- Swapping two columns multiplies  $\delta$  by  $-1$ .
- $\delta(e_1, \dots, e_d) = 1$ .

Let  $a_{ij}$  denote the entry in row  $i$  and column  $j$  of  $A$ , and let  $A^{(i,j)}$  be the matrix obtained by deleting row  $i$  and column  $j$ . The entrywise derivative of the determinant is

$$\partial_{a_{ij}} \det(A) = (-1)^{i+j} \det(A^{(i,j)}).$$

The quantity  $(-1)^{i+j} \det(A^{(i,j)})$  is the *cofactor* corresponding to the entry  $a_{ij}$ .

*Proof.* Expand the determinant along column  $j$ :

$$\det(A) = \sum_{r=1}^d (-1)^{r+j} a_{rj} \det(A^{(r,j)}).$$

Differentiating with respect to  $a_{ij}$  gives

$$\begin{aligned} \partial_{a_{ij}} \det(A) &= \partial_{a_{ij}} \left[ \sum_{r=1}^d (-1)^{r+j} a_{rj} \det(A^{(r,j)}) \right] \\ &= (-1)^{i+j} \det(A^{(i,j)}). \end{aligned}$$

Only the term with  $r = i$  contains  $a_{ij}$  directly. □

### 0.2.2.10 Higher-Order Derivatives

Higher-order derivatives are defined iteratively from first-order derivatives. For  $f: U \rightarrow \mathbb{R}^m$ ,

$$\partial_{i_k} \cdots \partial_{i_1} f: U \rightarrow \mathbb{R}^m,$$

where

$$x \mapsto \partial_{i_k} (\partial_{i_{k-1}} \cdots \partial_{i_1} f)(x).$$

The order of the indices records the order in which the derivatives are taken. Under suitable smoothness conditions, mixed partial derivatives commute.

**0.2.2.11 Hessian**

For a scalar-valued function  $f: U \rightarrow \mathbb{R}$ , the *Hessian* is

$$\nabla^2 f(x) := \begin{pmatrix} \partial_1 \partial_1 f(x) & \cdots & \partial_1 \partial_d f(x) \\ \vdots & \ddots & \vdots \\ \partial_d \partial_1 f(x) & \cdots & \partial_d \partial_d f(x) \end{pmatrix}.$$

The Hessian records the second-order curvature of a scalar-valued function.

**Example 0.2.3** (Computing a Hessian). Let

$$f(x^{(1)}, x^{(2)}) = (x^{(1)})^2 + 3x^{(1)}x^{(2)} + 2(x^{(2)})^2.$$

The first derivatives are

$$\partial_1 f = 2x^{(1)} + 3x^{(2)}, \quad \partial_2 f = 3x^{(1)} + 4x^{(2)}.$$

Therefore, the Hessian is

$$\nabla^2 f(x) = \begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}.$$

**0.2.2.12 Laplacian**

For a scalar-valued function  $f$ , the *Laplacian* is

$$\Delta f := \text{tr}(\nabla^2 f).$$

Equivalently,

$$\Delta f = \partial_1^2 f + \partial_2^2 f + \cdots + \partial_d^2 f,$$

where  $\partial_j^2 f := \partial_j \partial_j f$ .

The Laplacian appears in diffusion, score models, graph Laplacians, manifold learning, and regularization.

**0.2.2.13 Taylor's Theorem****0.2.2.14 One-Dimensional Case**

Let  $f: I \subset \mathbb{R} \rightarrow \mathbb{R}$  and suppose  $x + h \in I$ . The order- $k$  Taylor expansion used in these notes is

$$f(x + h) = f(x) + \partial f(x)h + \frac{1}{2}\partial^2 f(x)h^2 + \cdots + \frac{1}{k!}\partial^k f(x)h^k + o(h^k),$$

where

$$\lim_{h \rightarrow 0} \frac{o(h^k)}{|h|^k} = 0.$$

Thus,  $o(h^k)$  denotes an error term that is negligible compared with  $|h|^k$  as  $h \rightarrow 0$ .

An alternative version gives an explicit remainder. There exists a point  $\xi \in [x, x + h]$  such that

$$f(x + h) = f(x) + \partial f(x)h + \cdots + \frac{1}{(k-1)!} \partial^{k-1} f(x)h^{k-1} + \frac{1}{k!} \partial^k f(\xi)h^k.$$

The derivatives through order  $k-1$  are evaluated at  $x$ , while the final remainder uses  $\partial^k f(\xi)$  for some  $\xi$  between  $x$  and  $x + h$ .

### 0.2.2.15 Second-Order Approximation

Taking  $k = 2$  in the one-dimensional remainder formula gives

$$f(x + h) = f(x) + \partial f(x)h + \frac{1}{2} \partial^2 f(\xi)h^2, \quad \xi \in [x, x + h].$$

For a high-dimensional loss, the analogous approximation is

$$L(w + h) \approx L(w) + dL(w)h + \frac{1}{2} h^T \nabla^2 L(w)h.$$

The first-order term suggests a descent direction, while the second-order term controls the error and step size.

### 0.2.2.16 Gradient Descent from Taylor's Theorem

Take  $h = -\eta \nabla L(w)$ . Substitution into the second-order approximation gives

$$\begin{aligned} L(w + h) &\approx L(w) + dL(w)h + \frac{1}{2} h^T \nabla^2 L(w)h \\ &= L(w) - \eta \|\nabla L(w)\|_2^2 + \frac{\eta^2}{2} \nabla L(w)^T \nabla^2 L(w) \nabla L(w). \end{aligned}$$

The term  $-\eta \|\nabla L(w)\|_2^2$  decreases the loss, while the order- $\eta^2$  term is the curvature penalty. Choosing  $\eta$  too large can therefore destroy descent.

## 0.2.3 Probability

### 0.2.3.1 Sample Spaces and Events

The *sample space*  $\Omega$  is the set of all possible outcomes, and an *event* is a subset  $A \subseteq \Omega$ . A probability is a function assigning values to events,

$$\mathbb{P}: \mathcal{P}(\Omega) \rightarrow [0, 1],$$

such that

1.  $\mathbb{P}(\Omega) = 1$ ;
2.  $0 \leq \mathbb{P}(A) \leq 1$  for every  $A \subseteq \Omega$ ;



3. if  $A_1, A_2, \dots \subseteq \Omega$  are pairwise disjoint, then

$$\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i).$$

The pair  $(\Omega, \mathbb{P})$  is a *probability space*. Intuitively, the sample space lists all possible outcomes, while an event collects some of those outcomes.

**Example 0.2.4** (One die roll). For one roll of a fair die,

$$\Omega = \{1, 2, 3, 4, 5, 6\}.$$

The event that the outcome is even is

$$A = \{2, 4, 6\}.$$

Since three of the six outcomes are even,

$$\mathbb{P}(A) = \frac{3}{6} = \frac{1}{2}.$$

### 0.2.3.2 Random Variables and Realizations

Random variables are functions defined on a probability space. We use capital letters  $X, Y, \dots$  for random variables and lowercase letters  $x, y, \dots$  for their realizations or observed values. The notation

$$[X = x]$$

denotes the event

$$\{\omega \in \Omega : X(\omega) = x\}.$$

In supervised learning, we often regard  $(X, Y)$  as a random input–output pair. A dataset

$$(x_1, y_1), \dots, (x_n, y_n)$$

consists of observed realizations of independent copies of  $(X, Y)$ . Thus, capital  $X, Y$  denote random variables before the data are observed, whereas lowercase  $x_i, y_i$  denote observed data points.

### 0.2.3.3 Inclusion–Exclusion

For events  $A_1, A_2 \subseteq \Omega$ ,

$$\mathbb{P}(A_1 \cup A_2) = \mathbb{P}(A_1) + \mathbb{P}(A_2) - \mathbb{P}(A_1 \cap A_2).$$

The intersection is included once in each of  $\mathbb{P}(A_1)$  and  $\mathbb{P}(A_2)$ , so it must be subtracted once. Induction generalizes this identity to any finite collection of events.

### 0.2.3.4 Discrete Random Variables

A discrete random variable takes values in a discrete or countable space and has a *probability mass function* (p.m.f.). For a fair coin flip represented by the values 1 and 2,

$$R_X = \{1, 2\}, \quad \mathbb{P}(X = 1) = \mathbb{P}(X = 2) = \frac{1}{2}.$$

For a Poisson random variable,

$$R_X = \mathbb{N}, \quad \mathbb{P}(X = i) = \frac{e^{-\lambda} \lambda^i}{i!}, \quad \lambda > 0.$$

### 0.2.3.5 Continuous Random Variables

A continuous random variable takes values in a subset of  $\mathbb{R}^d$  and has a *probability density function* (p.d.f.). The uniform density on  $[0, 1]$  is

$$f(x) = \begin{cases} 1, & x \in [0, 1], \\ 0, & \text{otherwise.} \end{cases}$$

The density of a Gaussian random variable  $X \sim \mathcal{N}(\mu, \sigma^2)$  is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

### 0.2.3.6 Expectation

Some random variables are neither purely discrete nor purely continuous. For a discrete random variable with p.m.f.  $f$ ,

$$\mathbb{E}[X] = \sum_{x \in R_X} x f(x).$$

For a continuous random variable with p.d.f.  $f$ ,

$$\mathbb{E}[X] = \int_{x \in R_X} x f(x) \, dx.$$

Expectation is a probability-weighted average.

In machine learning, the population risk is an expectation:

$$R(f) = \mathbb{E}[\ell(f(X), Y)].$$

Usually, however, we observe only the data

$$(x_1, y_1), \dots, (x_n, y_n).$$

We therefore approximate the population risk by the empirical risk

$$\hat{R}(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i).$$

Empirical risk minimization replaces an unknown expectation with a computable average.

### 0.2.3.7 Convexity and Jensen's Inequality

A function  $\varphi$  is *convex* if, for every  $t \in [0, 1]$  and every  $x_1, x_2$  in its domain,

$$\varphi(tx_1 + (1-t)x_2) \leq t\varphi(x_1) + (1-t)\varphi(x_2).$$

Thus, the graph of a convex function lies below the line segment joining any two points on its graph. For an affine function, expectation commutes with the function:

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

For a nonlinear function  $\varphi$ , however,  $\mathbb{E}[\varphi(X)]$  and  $\varphi(\mathbb{E}[X])$  need not be equal.

For convex  $\varphi$ , *Jensen's inequality* states that

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

The left side first averages  $X$  and then applies  $\varphi$ ; the right side first applies  $\varphi$  and then averages. Convexity says the latter is at least as large. Jensen's inequality appears in likelihood bounds, variational inference, expectation-maximization, and many generalization and optimization arguments.

**Example 0.2.5** (A ratio of identically distributed variables). Suppose  $X, Y > 0$  have the same distribution and the relevant expectations exist. Because  $\log$  is concave, Jensen's inequality applied to  $X/Y$  gives

$$\log \mathbb{E} \left[ \frac{X}{Y} \right] \geq \mathbb{E} \left[ \log \frac{X}{Y} \right].$$

Since  $X$  and  $Y$  have the same distribution,

$$\mathbb{E} \left[ \log \frac{X}{Y} \right] = \mathbb{E}[\log X] - \mathbb{E}[\log Y] = 0.$$

Consequently,

$$\mathbb{E} \left[ \frac{X}{Y} \right] \geq 1.$$

This argument does not require  $X$  and  $Y$  to be independent.

### 0.2.3.8 Conditional Probability and Bayes' Rule

Let  $X: \Omega \rightarrow R_X$  and  $Y: \Omega \rightarrow R_Y$  be discrete random variables. If  $\mathbb{P}(X = x) > 0$ , define

$$\mathbb{P}(Y = y \mid X = x) = \frac{\mathbb{P}(Y = y, X = x)}{\mathbb{P}(X = x)}.$$

If  $\mathbb{P}(X = x) = 0$ , this expression is undefined in the elementary discrete setting.

For a fixed  $x$  with  $\mathbb{P}(X = x) > 0$ , conditional probability is a p.m.f. in  $y$  because

$$\begin{aligned} \sum_{y \in R_Y} \mathbb{P}(Y = y \mid X = x) &= \sum_{y \in R_Y} \frac{\mathbb{P}(Y = y, X = x)}{\mathbb{P}(X = x)} \\ &= \frac{\sum_{y \in R_Y} \mathbb{P}(Y = y, X = x)}{\mathbb{P}(X = x)} \\ &= \frac{\mathbb{P}(X = x)}{\mathbb{P}(X = x)} = 1. \end{aligned}$$

In both discrete and continuous settings, Bayes' rule has the form

$$\mathbb{P}(Y \mid X) = \frac{\mathbb{P}(X \mid Y)\mathbb{P}(Y)}{\mathbb{P}(X)},$$

with probabilities replaced by densities when appropriate. In the discrete case,

$$\mathbb{P}(Y = y \mid X = x) = \frac{\mathbb{P}(X = x \mid Y = y)\mathbb{P}(Y = y)}{\mathbb{P}(X = x)}.$$

Bayes classifiers choose the class having the largest posterior probability.

**Example 0.2.6** (A diagnostic test). Let  $D$  be the event that a person has a rare disease and  $A$  the event of a positive test. Suppose

$$\mathbb{P}(D) = 0.01, \quad \mathbb{P}(A \mid D) = 0.95, \quad \mathbb{P}(A \mid D^c) = 0.03.$$

Bayes' rule gives

$$\begin{aligned} \mathbb{P}(D \mid A) &= \frac{\mathbb{P}(A \mid D)\mathbb{P}(D)}{\mathbb{P}(A \mid D)\mathbb{P}(D) + \mathbb{P}(A \mid D^c)\mathbb{P}(D^c)} \\ &= \frac{0.95(0.01)}{0.95(0.01) + 0.03(0.99)} \approx 0.242. \end{aligned}$$

### 0.2.3.9 Conditional Expectation

Since  $\mathbb{P}(Y = y \mid X = x)$  is a p.m.f. in  $y$ , define

$$\mathbb{E}[Y \mid X = x] = \sum_{y \in R_Y} y \mathbb{P}(Y = y \mid X = x), \quad \mathbb{P}(X = x) > 0.$$

For continuous random variables, the corresponding definition replaces the sum with an integral. The quantity  $\mathbb{E}[Y \mid X = x]$  is the average value of  $Y$  after restricting to the slice where  $X = x$ .

Conditional expectation given  $X$  depends only on  $X$ . For each possible value  $x$ , define

$$g(x) := \mathbb{E}[Y \mid X = x].$$

Then

$$\mathbb{E}[Y \mid X] = g(X).$$

For squared-loss regression, the Bayes regressor is

$$x \mapsto \mathbb{E}[Y \mid X = x].$$

If many data points have features close to  $x$ , the best squared-loss prediction is their conditional average response. Many regression methods seek to approximate this unknown function from data.

### 0.2.3.10 Law of Total Expectation

The *law of total expectation*, also called the tower property, states that

$$\mathbb{E}[\mathbb{E}[Y \mid X]] = \mathbb{E}[Y].$$

The intuition is to condition on  $X$ , average  $Y$  within each slice  $X = x$ , and then average over all slices. This identity is used when decomposing expected risk, analyzing algorithms, and proving expected running-time results.

For discrete random variables,

$$\begin{aligned} \mathbb{E}[\mathbb{E}[Y \mid X]] &= \sum_{x \in R_X} \mathbb{E}[Y \mid X = x] \mathbb{P}(X = x) \\ &= \sum_{x \in R_X} \left( \sum_{y \in R_Y} y \mathbb{P}(Y = y \mid X = x) \right) \mathbb{P}(X = x) \\ &= \sum_{x \in R_X} \sum_{y \in R_Y} y \mathbb{P}(Y = y, X = x) \\ &= \sum_{y \in R_Y} y \mathbb{P}(Y = y) \\ &= \mathbb{E}[Y]. \end{aligned}$$

**Example 0.2.7** (Ten coins with different head probabilities). Independently flip ten coins, where coin  $i$  has probability  $1/i$  of landing heads. Let  $A$  be the event that the total number of heads is even, and let  $B$  be the event that the number of heads among coins  $1, 3, 4, \dots, 10$  is even. Define the indicator variables

$$Y = \mathbb{1}_A, \quad X = \mathbb{1}_B.$$

Then  $\mathbb{E}[Y] = \mathbb{P}(A)$ . Conditional on  $B$ , the total number of heads is even exactly when coin 2 lands tails. Conditional on  $B^c$ , it is even exactly when coin 2 lands heads. Since coin 2 is fair and independent of the remaining coins,

$$\mathbb{E}[Y \mid X = 1] = \mathbb{P}(A \mid B) = \frac{1}{2}, \quad \mathbb{E}[Y \mid X = 0] = \mathbb{P}(A \mid B^c) = \frac{1}{2}.$$

Therefore  $\mathbb{E}[Y \mid X] = 1/2$ , and the law of total expectation gives

$$\mathbb{P}(A) = \mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]] = \frac{1}{2}.$$

## 0.2.4 Matrix Analysis

### 0.2.4.1 Matrix Notation

For real matrices, write

$$\mathcal{M}_{n,d}(\mathbb{R}) = \{n \times d \text{ real matrices}\}, \quad \mathcal{M}_d(\mathbb{R}) = \mathcal{M}_{d,d}(\mathbb{R}).$$

The set of  $d \times d$  orthogonal matrices is

$$\mathcal{O}_d(\mathbb{R}) = \{U \in \mathcal{M}_d(\mathbb{R}) : U^T U = U U^T = I_d\}.$$

The columns of an orthogonal matrix form an orthonormal basis.

### 0.2.4.2 Eigen-Decomposition of Symmetric Matrices

Let  $A \in \mathcal{M}_d(\mathbb{R})$  be symmetric, so  $A = A^T$ . Then there exist an orthogonal matrix  $U \in \mathcal{O}_d(\mathbb{R})$  and a diagonal matrix  $D \in \mathcal{M}_d(\mathbb{R})$  such that

$$A = U D U^T.$$

The diagonal entries of  $D$  are the eigenvalues of  $A$ , and the columns of  $U$  are orthonormal eigenvectors of  $A$ .

More explicitly, write

$$U = \begin{bmatrix} u_1 & \cdots & u_d \end{bmatrix}, \quad D = \text{diag}(\lambda_1, \dots, \lambda_d).$$

Then  $A = U D U^T$  means

$$A u_i = \lambda_i u_i, \quad i = 1, \dots, d,$$

and orthonormality gives

$$u_i^T u_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Geometrically, a symmetric matrix stretches or compresses along mutually orthogonal eigendirections.

**Example 0.2.8** (A diagonal symmetric matrix). Let

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then

$$A = U D U^T, \quad U = I_2, \quad D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

The eigenvectors are

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

with eigenvalues 3 and 1, respectively.

Eigen-decomposition is foundational for:

- principal component analysis, through the variance directions of covariance matrices;
- spectral clustering, through eigenvectors of graph Laplacians;
- optimization, through Hessian eigenvalues and curvature;
- Gaussian models, through covariance structure;
- regularization and conditioning.

The main idea is that a symmetric matrix becomes simple after rotating to its eigenbasis.

### 0.2.4.3 Positive Semidefinite Matrices

A symmetric matrix  $A \in \mathcal{M}_d(\mathbb{R})$  is *positive semidefinite*, written  $A \succeq 0$ , if

$$x^T A x \geq 0 \quad \text{for every } x \in \mathbb{R}^d.$$

If

$$A = U \operatorname{diag}(\lambda_1, \dots, \lambda_d) U^T,$$

then the equivalent eigenvalue characterization is

$$A \succeq 0 \iff \lambda_i \geq 0 \text{ for every } i.$$

### 0.2.4.4 Matrix Square Root

Let  $A \succeq 0$  have eigen-decomposition

$$A = U \operatorname{diag}(\lambda_1, \dots, \lambda_d) U^T, \quad \lambda_i \geq 0.$$

Define its square root by

$$A^{1/2} := U \operatorname{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_d}) U^T.$$

Then

$$A^{1/2} A^{1/2} = A.$$

Covariance matrices are positive semidefinite, and their square roots appear in Gaussian sampling and whitening transformations.

### 0.2.4.5 Singular Value Decomposition

Let  $A \in \mathcal{M}_{n,d}(\mathbb{R})$ . Then there exist orthogonal matrices

$$U \in \mathcal{O}_n(\mathbb{R}), \quad V \in \mathcal{O}_d(\mathbb{R}),$$

and a rectangular diagonal matrix  $D \in \mathcal{M}_{n,d}(\mathbb{R})$  with nonnegative diagonal entries such that

$$A = U D V^T.$$

The diagonal entries of  $D$  are the *singular values* of  $A$ .

Let  $m = \min(n, d)$  and order the singular values as

$$\sigma_1(A) \geq \sigma_2(A) \geq \cdots \geq \sigma_m(A) \geq 0.$$

Writing  $u_i$  and  $v_i$  for the corresponding left and right singular vectors, respectively, the SVD can also be read as the rank-one expansion

$$A = \sum_{i=1}^m \sigma_i(A) u_i v_i^T.$$

Geometrically, a linear map first rotates, then stretches along orthogonal directions by its singular values, and then rotates again. In two dimensions, the image of the unit circle is an ellipse whose principal semiaxes point along the left singular vectors and have lengths given by the singular values.

**Reduced SVD** Let  $r = \text{rank}(A)$ . The reduced SVD retains only the nonzero singular values:

$$A = U_r D_r V_r^T,$$

where

$$U_r \in \mathbb{R}^{n \times r}, \quad D_r = \text{diag}(\sigma_1, \dots, \sigma_r) \in \mathbb{R}^{r \times r}, \quad V_r \in \mathbb{R}^{d \times r}.$$

It stores only the active directions and is therefore more efficient when  $r \ll \min(n, d)$ .

The SVD is foundational for:

- principal component analysis and dimensionality reduction;
- spectral clustering;
- conditioning and numerical stability;
- matrix completion and recommender systems;
- model compression and low-rank regularization.

Large singular values identify directions in which the matrix acts strongly, while small singular values identify weak or noisy directions.